

# The cylindric Lorentzian problem at $\ell = 3$ : a reduction to two combinatorial inequalities

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29 September 2026

## Abstract

Let  $s_{\lambda/\mu}^c$  be the cylindric skew Schur polynomial of a cylindric shape at level  $n$  with  $m$  beads, and let  $N(s^c) = \sum_{\alpha} (K_{\alpha}^c / \alpha!) x^{\alpha}$  be its Brändén–Huh normalisation. The open regime of the conjecture “ $N(s^c)$  is Lorentzian” is  $\ell \geq 3$  and  $m \geq 2$ . This note does not prove the conjecture at  $\ell = 3$ . It does three things. First, it *reduces* the  $\ell = 3$  case to two explicit combinatorial inequalities on the coefficient array, by proving an elementary criterion (Lemma 3) for a nonnegative symmetric  $3 \times 3$  matrix to have at most one positive eigenvalue: conditions (A)  $M_{ij}^2 \geq M_{ii}M_{jj}$  and (B)  $M_{ik}M_{kj} \geq M_{ij}M_{kk}$ . Together these say that  $\phi_{ij} = \log(M_{ij}/\sqrt{M_{ii}M_{jj}})$  is a pseudometric. Condition (B) is new; it is precisely what the previously recorded dead end `rlc-implies-13` was missing, and it is violated by that dead end’s own witness. Second, it establishes the *scope* of the criterion: Lemma 3 is strictly a  $3 \times 3$  phenomenon, failing at  $\ell = 4$  and, structurally, at  $\ell = 5$  via the  $K_{2,3}$  graph metric. Third, it proves the two inequalities in the totally ordered case (Lemmas 12 and 13) and isolates the remaining gap to a single precisely stated failure: monotone likelihood ratio holds across *comparable* inner shapes but genuinely fails across *incomparable* ones, while the fibrewise sums nevertheless satisfy it. Conditions (A) and (B) are verified with exact integer arithmetic on 220,658 Hessians.

## 1 Setting

### 1.1 Cylindric shapes and cylindric Schur polynomials

Fix integers  $n > m \geq 1$ . A *cylindric shape* at level  $n$  with  $m$  beads is a bi-infinite strictly increasing sequence  $(x_i)_{i \in \mathbb{Z}}$  with  $x_{i+m} = x_i + n$ ; we store it as the tuple  $x = (x_1, \dots, x_m)$  with  $x_1 < \dots < x_m < x_1 + n$ , and we always read indices cyclically through  $x_{i+m} = x_i + n$ . Write  $\nu \subseteq \rho$  for  $x_i(\nu) \leq x_i(\rho)$  for all  $i$ , and  $|\rho/\nu| = \sum_{i=1}^m (x_i(\rho) - x_i(\nu))$ .

The pair  $\rho/\nu$  is a *cylindric horizontal strip* when

$$\nu_i \leq \rho_i < \nu_{i+1} \quad (1 \leq i \leq m), \tag{1}$$

with  $\nu_{m+1} = \nu_1 + n$ . A *cylindric SSYT* of shape  $\lambda/\mu$  with  $\ell$  rows is a chain

$$\mu = \kappa^0 \subseteq \kappa^1 \subseteq \dots \subseteq \kappa^{\ell} = \lambda$$

in which every  $\kappa^t/\kappa^{t-1}$  is a cylindric horizontal strip; its weight is  $\alpha = (\alpha_1, \dots, \alpha_{\ell})$  with  $\alpha_t = |\kappa^t/\kappa^{t-1}|$ . Set

$$K_{\alpha}^c = \#\{\text{cylindric SSYT of shape } \lambda/\mu \text{ and weight } \alpha\}, \quad s_{\lambda/\mu}^c(x_1, \dots, x_{\ell}) = \sum_{\alpha} K_{\alpha}^c x^{\alpha}.$$

This is homogeneous of degree  $d = |\lambda/\mu|$  and is a symmetric polynomial; in particular  $K_{\alpha}^c = K_{w\alpha}^c$  for every permutation  $w$  of the  $\ell$  coordinates. That symmetry is used repeatedly below and is not re-proved here.

## 1.2 The normalisation and the three conditions

Put  $N(s^c) = \sum_{\alpha} \frac{K_{\alpha}^c}{\alpha!} x^{\alpha}$ . The conditions defining Lorentzian-ness in the form used throughout this programme are

- (L1) all coefficients are nonnegative — immediate, since  $K_{\alpha}^c$  counts tableaux;
- (L2)  $\text{supp}(s^c)$  is M-convex — already **proved** and Lean-verified in this programme (registry `cylindric-M-convex` `SortedBridge.sorted_exchange`, sorry-free);
- (L3) for every  $\beta \in \mathbb{N}^{\ell}$  with  $|\beta| = d - 2$ , the quadratic form  $\partial^{\beta} N(s^c)$  has at most one positive eigenvalue.

By the recorded **raw-hessian-lemma** (**proved**, and re-checked symbolically in this session), the matrix of the quadratic form  $\partial^{\beta} N(s^c)$  is

$$M_{ij}^{(\beta)} = K_{\beta+e_i+e_j}^c, \quad (2)$$

the *raw* coefficients, not the normalised ones. All matrices below are the raw ones.

Throughout,  $M = M^{(\beta)}$  is symmetric with nonnegative integer entries, and  $\text{tr } M \geq 0$ .

## 2 The two conditions and the criterion

**Definition 1.** Let  $M$  be a symmetric  $\ell \times \ell$  matrix with nonnegative entries. Say  $M$  satisfies

$$M_{ij}^2 \geq M_{ii}M_{jj} \quad \text{for all } i \neq j, \quad (\text{A})$$

$$M_{ik}M_{kj} \geq M_{ij}M_{kk} \quad \text{for all distinct } i, j, k. \quad (\text{B})$$

**Remark 2** (Why these two). If every  $M_{ii} > 0$ , set  $N_{ij} = M_{ij}/\sqrt{M_{ii}M_{jj}}$ , so that  $N$  is congruent to  $M$  and  $N_{ii} = 1$ . Then (A) says  $N_{ij} \geq 1$  and (B) says  $N_{ij} \leq N_{ik}N_{kj}$ . Writing  $\phi_{ij} = \log N_{ij}$ , the pair (A)+(B) says exactly that  $\phi$  is a *pseudometric*: nonnegative, symmetric, vanishing on the diagonal, and satisfying the triangle inequality. This is the shortest description of the criterion, and it is the one that explains its scope (§2.2). The proof below is written without dividing by diagonal entries, so that the degenerate cases  $M_{ii} = 0$  need no separate treatment.

In terms of the coefficients, with  $\alpha = \beta + e_i + e_j$ , condition (A) is

$$(K_{\alpha}^c)^2 \geq K_{\alpha+e_i-e_j}^c K_{\alpha-e_i+e_j}^c,$$

which is exactly the raw root-log-concavity (RLC) already instrumented in this programme. Condition (B) is new.

**Lemma 3** (Metric criterion). *Let  $M$  be a symmetric  $3 \times 3$  matrix with nonnegative real entries satisfying (A) and (B). Then  $e_2(M) \leq 0$ ,  $\det M \geq 0$ , and  $M$  has at most one positive eigenvalue. Moreover  $\det M = 0$  if and only if at least two of the three inequalities (B1)–(B3) below are equalities.*

*Proof.* Write

$$P = M_{11}M_{22}M_{33}, \quad t = M_{12}M_{13}M_{23}, \quad x = M_{11}M_{23}^2, \quad y = M_{22}M_{13}^2, \quad z = M_{33}M_{12}^2.$$

Expanding the determinant of a symmetric  $3 \times 3$  matrix gives the identity

$$\det M = P + 2t - x - y - z, \quad (3)$$

and directly from the definitions,

$$xyz = Pt^2. \quad (4)$$

*Step 1: (B) gives  $x, y, z \leq t$ .* Taking (B) with  $k = 1$ ,  $(i, j) = (2, 3)$  reads  $M_{21}M_{13} \geq M_{23}M_{11}$ ; multiplying both sides by  $M_{23} \geq 0$ ,

$$t = M_{12}M_{13}M_{23} \geq M_{11}M_{23}^2 = x. \quad (B1)$$

Taking  $k = 2$ ,  $(i, j) = (1, 3)$  reads  $M_{12}M_{23} \geq M_{13}M_{22}$ ; multiplying by  $M_{13} \geq 0$ ,

$$t \geq M_{22}M_{13}^2 = y. \quad (B2)$$

Taking  $k = 3$ ,  $(i, j) = (1, 2)$  reads  $M_{13}M_{32} \geq M_{12}M_{33}$ ; multiplying by  $M_{12} \geq 0$ ,

$$t \geq M_{33}M_{12}^2 = z. \quad (B3)$$

*Step 2:  $\det M \geq 0$ .* Suppose first  $t = 0$ . Then (B1)–(B3) force  $x = y = z = 0$ , and (A) multiplied out gives  $t^2 = M_{12}^2M_{13}^2M_{23}^2 \geq (M_{11}M_{22})(M_{11}M_{33})(M_{22}M_{33}) = P^2$ , so  $P = 0$ . By (3),  $\det M = 0$ .

Now suppose  $t > 0$ . By (4),  $P = xyz/t^2$ , so (3) gives

$$t^2 \det M = xyz + 2t^3 - t^2(x + y + z). \quad (5)$$

The right-hand side is nonnegative on  $0 \leq x, y, z \leq t$  because of the polynomial identity

$$xyz + 2t^3 - t^2(x + y + z) = x(t - y)(t - z) + t(t - x)(t - y) + t(t - x)(t - z), \quad (6)$$

whose three summands are each a product of nonnegative factors, by Step 1 and nonnegativity of the entries. Dividing (5) by  $t^2 > 0$  gives  $\det M \geq 0$ .

For the equality condition,  $\det M = 0$  forces all three summands of (6) to vanish. The last two give  $(t - x)(t - y) = 0$  and  $(t - x)(t - z) = 0$ . If  $x = t$  the first summand is  $t(t - y)(t - z) = 0$ , so  $y = t$  or  $z = t$ ; if  $x \neq t$  then  $y = t$  and  $z = t$ . Either way at least two of  $x, y, z$  equal  $t$ . Conversely, if at least two of them equal  $t$  then each summand of (6) vanishes. By (B1)–(B3) this is exactly the condition that at least two of those three inequalities are equalities.

*Remark on the shape of this step.* (6) is an explicit certificate: it replaces an appeal to “ $g = \xi\eta\zeta - \xi - \eta - \zeta + 2$  is affine in each variable, hence minimised at a vertex of  $[0, 1]^3$ ”, which is how the result was first found. That vertex argument is true and is a genuine lemma, but it is *not needed* — a fact discovered only when the step was formalised (Remark 5). Where the vertex argument quantifies over extreme points and needs the rescaling  $\xi = x/t$ , (6) is a single **ring** identity in  $x, y, z, t$  followed by three products of nonnegative factors, and it yields the equality condition directly rather than by a separate affineness argument.

*Step 3:  $e_2(M) \leq 0$ .* The second elementary symmetric function of the eigenvalues equals the sum of the three  $2 \times 2$  principal minors,

$$e_2(M) = \sum_{i < j} (M_{ii}M_{jj} - M_{ij}^2),$$

and each summand is  $\leq 0$  by (A).

*Step 4: conclusion.* Let  $\lambda_1 \geq \lambda_2 \geq \lambda_3$  be the eigenvalues and suppose  $\lambda_2 > 0$ . Then  $\lambda_1 \geq \lambda_2 > 0$ , and  $\det M = \lambda_1\lambda_2\lambda_3 \geq 0$  forces  $\lambda_3 \geq 0$ . Hence  $e_2(M) = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 \geq \lambda_1\lambda_2 > 0$ , contradicting Step 3. So  $\lambda_2 \leq 0$ .  $\square$

**Remark 4.** Steps 1–2 use only (B) (with (A) needed solely in the degenerate case  $t = 0$ ), and Step 3 uses only (A). The roles are cleanly separated: (A) controls  $e_2$ , (B) controls the determinant.

**Remark 5** (Formalised, and simplified in the process). Lemma 3 and Proposition 6 are **verified in Lean 4/Mathlib** as `TworowD4Kernel.MetricCriterion.metric_criterion` and `...l3.det_reduction` (repository `clio-vega/tworow-d4-kernel`), sorry-free, with axioms exactly `[propext, Classical.choice, Quot.sound]`. The eigenvalues there are Mathlib’s `Matrix.IsHermitian.eigenvalues`, and the identification of  $\text{tr}$ ,  $e_2$ ,  $\det$  with the elementary symmetric functions of the spectrum is derived from the characteristic polynomial rather than assumed.

The formalisation *removed* a step, and Step 2 above is now written in the form that was actually checked: the certificate (6), one `ring` identity plus three applications of `mul_nonneg`. The superseded appeal to “ $g$  is affine in each variable, hence minimised at a vertex of  $[0, 1]^3$ ” has been removed from the proof. It was true, and a real lemma; it was simply not load-bearing — which no amount of checking whether the steps are *correct* would have revealed, since a true step is invisible to a falsification sweep by construction. The  $t = 0$  branch of Step 2 is unaffected: (6) holds identically at  $t = 0$  too, but the passage from (5) to  $\det M \geq 0$  needs  $t > 0$ , so that branch is still argued separately (and was already present and sound).

The converse direction of Step 4 also holds and is worth recording, because it says that at  $\ell = 3$  nothing is lost by the reformulation.

**Proposition 6** (Reduction). *Let  $M$  be a symmetric  $3 \times 3$  matrix with nonnegative entries, so that  $\text{tr } M \geq 0$ . Then  $M$  has at most one positive eigenvalue if and only if  $e_2(M) \leq 0$  and  $\det M \geq 0$ .*

*Proof.*  $(\Leftarrow)$  is Step 4 above.  $(\Rightarrow)$  Let  $\lambda_1 \geq 0 \geq \lambda_2 \geq \lambda_3$ . Then  $\det M = \lambda_1 \lambda_2 \lambda_3 \geq 0$ . Moreover  $\text{tr } M \geq 0$  gives  $\lambda_1 \geq -(\lambda_2 + \lambda_3) \geq 0$ , and since  $\lambda_2 + \lambda_3 \leq 0$ ,

$$e_2(M) = \lambda_1(\lambda_2 + \lambda_3) + \lambda_2 \lambda_3 \leq -(\lambda_2 + \lambda_3)^2 + \lambda_2 \lambda_3 = -(\lambda_2^2 + \lambda_2 \lambda_3 + \lambda_3^2) \leq 0. \quad \square$$

**Corollary 7.** *At  $\ell = 3$ , condition (L3) for a given  $\beta$  is equivalent to the conjunction of RLC at  $\beta$  and  $\det M^{(\beta)} \geq 0$ . Since RLC has been verified without exception throughout this programme, **the entire open content of (L3) at  $\ell = 3$  is the single determinantal inequality  $\det M^{(\beta)} \geq 0$ .***

## 2.1 Condition (B) is exactly what the recorded dead end was missing

The registry node `rlc-implies-l3` is a **dead-end** whose witness is

$$W = \begin{pmatrix} 4 & 6 & 4 \\ 6 & 1 & 4 \\ 4 & 4 & 4 \end{pmatrix},$$

a strictly positive symmetric matrix satisfying every  $2 \times 2$  condition ( $36 \geq 4$ ,  $16 \geq 16$ ,  $16 \geq 4$ ) yet having characteristic polynomial  $x^3 - 9x^2 - 44x + 16$  and therefore two positive eigenvalues. In the present language:  $W$  satisfies (A) and  $\det W = -16 < 0$ . Checking (B) at  $(i, j, k) = (1, 2, 3)$ :  $W_{13}W_{32} = 16$  while  $W_{12}W_{33} = 24$ , so (B) *fails*. Equivalently, in the normalisation of Remark 2,  $(N_{12}, N_{13}, N_{23}) = (3, 1, 2)$  and the triangle inequality  $N_{12} \leq N_{13}N_{23} = 2$  is violated. Condition (B) is thus not an arbitrary strengthening: it is precisely the hypothesis whose absence produced the recorded counterexample.

## 2.2 Scope of Lemma 3

It matters, for this programme, to record not only that a reason is true but for which quantifiers. Lemma 3 is *strictly a  $3 \times 3$  statement*.

**Proposition 8.** *For  $\ell \geq 4$  there exist symmetric  $\ell \times \ell$  matrices with nonnegative entries satisfying (A) and (B) and having two positive eigenvalues.*

*Proof and structural witness.* A random integer search at  $\ell = 4$  found 777 such matrices among 4012 satisfying (A) and (B), for example  $\begin{pmatrix} 3 & 6 & 2 & 6 \\ 6 & 0 & 1 & 4 \\ 2 & 1 & 0 & 4 \\ 6 & 4 & 4 & 1 \end{pmatrix}$ . The conceptual reason appears at  $\ell = 5$ .

Suppose all  $M_{ii} > 0$  and normalise as in Remark 2, so  $N_{ij} = e^{\phi_{ij}}$  with  $\phi$  a pseudometric. On the hyperplane  $\{\sum_i x_i = 0\}$  one computes, using  $N_{ii} = 1$ ,

$$x^\top N x = \sum_i x_i^2 + \sum_{i \neq j} e^{\phi_{ij}} x_i x_j = \left( \sum_i x_i \right)^2 + \sum_{i \neq j} (e^{\phi_{ij}} - 1) x_i x_j = \sum_{i \neq j} (e^{\phi_{ij}} - 1) x_i x_j.$$

So  $N$  has at most one positive eigenvalue as soon as the nonnegative hollow matrix  $(e^{\phi_{ij}} - 1)$  is conditionally negative definite, i.e. “of negative type”. Not every metric is of negative type, and the smallest classical obstruction is the shortest-path metric of  $K_{2,3}$ . Taking  $\phi$  to be that metric on five points ( $\phi = 1$  across the parts,  $\phi = 2$  within each part) and  $N_{ij} = e^{\phi_{ij}}$ , conditions (A) and (B) hold, while the spectrum is  $\{-6.389, -6.389, -6.389, 4.469, 19.698\}$ : two positive eigenvalues. An exact integer rescaling confirms this with rational arithmetic.  $\square$

Proposition 8 does *not* refute (L3) for cylindric shapes at  $\ell \geq 4$ ; the cylindric pseudometrics may all be of negative type. It says only that the criterion used here cannot be the mechanism beyond  $\ell = 3$ . This is exactly why  $\ell = 3$  was the right target, and the reason is now explained rather than merely chosen.

## 3 The two inequalities as combinatorics

Because  $s^c$  is symmetric, the three instances of (A) are equivalent to one another, and likewise the three instances of (B). Writing  $c = d - a - b$  and

$$k(a, b) := K_{(a, b, d-a-b)}^c,$$

the six Hessian inequalities collapse to two statements about the triangular array  $k$ .

**Proposition 9.** *At  $\ell = 3$ , condition (A) holds for all  $\beta$  if and only if  $k$  is log-concave in  $a$ ,*

$$k(a, b)^2 \geq k(a+1, b) k(a-1, b),$$

*and condition (B) holds for all  $\beta$  if and only if  $k$  is log-submodular,*

$$k(a+1, b) k(a, b+1) \geq k(a, b) k(a+1, b+1).$$

*Verification.* Place  $\beta = (p, q, r)$  and let  $A = \beta + 2e_1$ ,  $B = \beta + 2e_2$ ,  $C = \beta + 2e_3$ ,  $F = \beta + e_1 + e_2$ ,  $G = \beta + e_1 + e_3$ ,  $H = \beta + e_2 + e_3$ , so that  $M = \begin{pmatrix} K_A & K_F & K_G \\ K_F & K_B & K_H \\ K_G & K_H & K_C \end{pmatrix}$ . Then (A) reads  $K_F^2 \geq K_A K_B$ ,  $K_G^2 \geq K_A K_C$ ,  $K_H^2 \geq K_B K_C$ , and (B) reads  $K_G K_H \geq K_F K_C$ ,  $K_F K_H \geq K_G K_B$ ,  $K_F K_G \geq K_H K_A$ . In  $(a, b)$  coordinates,  $A = (p+2, q)$ ,  $B = (p, q+2)$ ,  $C = (p, q)$ ,  $F = (p+1, q+1)$ ,  $G = (p+1, q)$ ,  $H = (p, q+1)$ ; substituting turns (A)<sub>13</sub> into log-concavity in  $a$  and (B) <sub>$k=3$</sub>  into log-submodularity, and the  $S_3$ -symmetry of  $K^c$  carries each family onto the displayed representative. The equivalence was additionally checked mechanically on all 573 instances with  $n \leq 6$ ,  $d \leq 9$ : zero mismatches in either direction.  $\square$

## 4 An interlacing model

The following description of  $K^c$  at  $\ell = 3$  is the foundation for the partial results of §5. It removes the chain-of-shapes formulation in favour of a single cyclically interlacing sequence with box constraints.

**Proposition 10.** *Fix a cylindric skew shape  $\lambda/\mu$  at level  $n$  with  $m$  beads. Then*

$$k(a, b) = \# \left\{ (\kappa, \sigma) \in \mathbb{Z}^m \times \mathbb{Z}^m : \begin{array}{l} \mu_i \leq \kappa_i \leq \mu_{i+1} - 1, \quad \lambda_{i-1} + 1 \leq \sigma_i \leq \lambda_i, \\ \kappa_i \leq \sigma_i \leq \kappa_{i+1} - 1 \quad (1 \leq i \leq m), \\ S_\kappa = |\mu| + a, \quad S_\sigma = |\mu| + a + b \end{array} \right\},$$

where  $S_\kappa = \sum_i \kappa_i$ , indices are read cyclically through  $x_{i+m} = x_i + n$ , and  $|\mu| = \sum_i \mu_i$ .

*Proof.* A cylindric SSYT with 3 rows is a chain  $\mu \subseteq \kappa \subseteq \sigma \subseteq \lambda$  of horizontal strips. By (1),  $\kappa/\mu$  is a strip iff  $\mu_i \leq \kappa_i < \mu_{i+1}$ ;  $\sigma/\kappa$  is a strip iff  $\kappa_i \leq \sigma_i < \kappa_{i+1}$ ; and  $\lambda/\sigma$  is a strip iff  $\sigma_i \leq \lambda_i$  and  $\lambda_i < \sigma_{i+1}$ , i.e.  $\lambda_{i-1} + 1 \leq \sigma_i \leq \lambda_i$ . No separate check that  $\kappa$  and  $\sigma$  are shapes is needed:  $\kappa_i \leq \mu_{i+1} - 1 < \mu_{i+1} \leq \kappa_{i+1}$  gives  $\kappa_i < \kappa_{i+1}$ , and likewise for  $\sigma$ . The weights are  $a = |\kappa/\mu| = S_\kappa - |\mu|$ ,  $b = |\sigma/\kappa| = S_\sigma - S_\kappa$ ,  $c = |\lambda/\sigma| = |\lambda| - S_\sigma$ .  $\square$

Proposition 10 was checked against the independent transfer-matrix dynamic program on all 911 instances with  $n \leq 6$ ,  $d \leq 8$ : zero disagreements.

Eliminating  $\kappa$  from the middle constraint exhibits  $(\kappa, \sigma)$  as a single cyclic chain

$$\kappa_1 \leq \sigma_1 < \kappa_2 \leq \sigma_2 < \cdots \leq \sigma_m < \kappa_1 + n,$$

so the constraint system consists solely of differences  $x_p - x_q \leq c$ . In particular the solution set  $\mathcal{L} \subseteq \mathbb{Z}^{2m}$  is closed under coordinatewise min and max, i.e. it is a sublattice of  $\mathbb{Z}^{2m}$ , and  $(S_\kappa, S_\sigma)$  is a linear — hence modular — functional on it. Writing  $P(u, v) = \#\{(\kappa, \sigma) \in \mathcal{L} : S_\kappa = u, S_\sigma = v\}$ , so that  $k(a, b) = P(|\mu| + a, |\mu| + a + b)$ , Proposition 9 says that (B) is exactly the statement that  $P$  is *log-supermodular* on  $\mathbb{Z}^2$ .

**Remark 11** (Why Ahlswede–Daykin does not apply directly). Log-supermodularity of a pushforward follows from the four functions theorem when the projection is a *lattice homomorphism*. Here it is not: for  $x, y \in \mathcal{L}$ ,  $S_{\kappa^x \vee \kappa^y} = \sum_i \max(\kappa_i^x, \kappa_i^y) \geq \max(S_{\kappa^x}, S_{\kappa^y})$ , and the inequality is generally strict. Modularity gives  $\pi(x \wedge y) + \pi(x \vee y) = \pi(x) + \pi(y)$  with  $\pi(x \vee y) \geq \pi(x) \vee \pi(y)$ , so the pair  $(x \wedge y, x \vee y)$  lands *outside* the fibres  $\pi(x) \wedge \pi(y)$  and  $\pi(x) \vee \pi(y)$  by a nonnegative slack. This obstruction is real and is the reason the argument of §5 takes a different route.

## 5 Partial results: the totally ordered case

Fix  $\kappa$  admissible. By Proposition 10, once  $\kappa$  is fixed each  $\sigma_i$  ranges *independently* over the interval  $[L_i(\kappa), R_i(\kappa)]$  with

$$L_i(\kappa) = \max(\kappa_i, \lambda_{i-1} + 1), \quad R_i(\kappa) = \min(\kappa_{i+1} - 1, \lambda_i).$$

Hence

$$P(u, v) = \sum_{\kappa : S_\kappa = u} W_\kappa(v), \quad W_\kappa = \mathbf{1}_{[L_1, R_1]} * \cdots * \mathbf{1}_{[L_m, R_m]}. \quad (7)$$

Each  $W_\kappa$  is a convolution of interval indicators, hence a Pólya frequency sequence of order 2 (PF<sub>2</sub>): log-concave with no internal zeros.

Recall that  $f \preceq_{\text{lr}} g$  (the *likelihood ratio* order, equivalently  $\text{TP}_2$  of the two-row matrix) means  $f(v)g(v') \geq f(v')g(v)$  for all  $v < v'$ ; and that a nonnegative sequence  $g$  is  $\text{PF}_2$  iff

$$g(p)g(p' + \delta) \geq g(p + \delta)g(p') \quad \text{for all } p > p', \delta \geq 0. \quad (8)$$

**Lemma 12** (Convolution preserves the likelihood ratio order). *Let  $f, f', g, g'$  be nonnegative  $\text{PF}_2$  sequences with  $f \preceq_{\text{lr}} f'$  and  $g \preceq_{\text{lr}} g'$ . Then  $f * g \preceq_{\text{lr}} f' * g'$ .*

*Proof.* By transitivity it suffices to prove  $f * g \preceq_{\text{lr}} f' * g$  for  $f \preceq_{\text{lr}} f'$  and  $g \text{ PF}_2$  (the second step,  $f' * g \preceq_{\text{lr}} f' * g'$ , is the same statement with the roles of the factors exchanged). Fix  $v < v'$  and set  $\delta = v' - v > 0$ . Then

$$S := (f * g)(v)(f' * g)(v') - (f * g)(v')(f' * g)(v) = \sum_{w, w'} f(w)f'(w') \Delta(w, w'),$$

where  $\Delta(w, w') = g(v - w)g(v' - w') - g(v' - w)g(v - w')$ . Exchanging the summation names  $w \leftrightarrow w'$  and using  $\Delta(w', w) = -\Delta(w, w')$  gives  $S = -\sum_{w, w'} f(w')f'(w)\Delta(w, w')$ , whence

$$2S = \sum_{w, w'} [f(w)f'(w') - f(w')f'(w)] \Delta(w, w').$$

Both brackets are antisymmetric in  $(w, w')$ . For  $w < w'$  the first is  $\geq 0$  because  $f \preceq_{\text{lr}} f'$ . For the second, put  $p = v - w$  and  $p' = v - w'$ , so  $p > p'$ ; then  $v' - w = p + \delta$  and  $v' - w' = p' + \delta$ , and

$$\Delta(w, w') = g(p)g(p' + \delta) - g(p + \delta)g(p') \geq 0$$

by (8). So for  $w < w'$  both factors are  $\geq 0$ , for  $w > w'$  both are  $\leq 0$ , and for  $w = w'$  the product vanishes. Every term of  $2S$  is therefore  $\geq 0$ , so  $S \geq 0$ .  $\square$

**Lemma 13** (Comparable inner shapes). *If  $\kappa \leq \kappa'$  coordinatewise and both are admissible, then  $W_\kappa \preceq_{\text{lr}} W_{\kappa'}$ .*

*Proof.*  $L_i$  is nondecreasing in  $\kappa_i$  and  $R_i$  is nondecreasing in  $\kappa_{i+1}$ , so  $L_i(\kappa) \leq L_i(\kappa')$  and  $R_i(\kappa) \leq R_i(\kappa')$  for every  $i$ . For two intervals with both endpoints weakly increasing,  $\mathbf{1}_{[L, R]} \preceq_{\text{lr}} \mathbf{1}_{[L', R']}$ : if  $v < v'$  and the right-hand side of the defining inequality is nonzero, then  $L' \leq v < v' \leq R$ , whence  $v \geq L' \geq L$  and  $v < v' \leq R$  give  $v \in [L, R]$ , and  $v' > v \geq L'$  with  $v' \leq R \leq R'$  give  $v' \in [L', R']$ , so the left-hand side is 1 as well. Interval indicators are  $\text{PF}_2$  and convolutions of  $\text{PF}_2$  sequences are  $\text{PF}_2$ , so Lemma 12 applies inductively over  $i = 1, \dots, m$ .  $\square$

**Corollary 14.** *Condition (B), hence (L3) together with RLC, holds at  $\ell = 3$  for every level  $n$  whenever the set of admissible inner shapes  $\kappa$  is a chain under containment — in particular for  $m = 1$ .*

*Proof.* If the admissible  $\kappa$  form a chain graded by  $S_\kappa$ , then for  $u < u'$  every  $\kappa$  in the fibre  $\{S_\kappa = u\}$  is  $\leq$  every  $\kappa'$  in  $\{S_{\kappa'} = u'\}$ . Summing a  $\text{TP}_2$  matrix over consecutive blocks of rows preserves  $\text{TP}_2$ , since each  $2 \times 2$  minor of the blocked matrix is a sum of  $2 \times 2$  minors of the original with rows taken from the earlier and the later block respectively. Hence  $P(u, \cdot) \preceq_{\text{lr}} P(u', \cdot)$  by (7) and Lemma 13, which is (B). At  $m = 1$  the shapes  $\kappa$  are single integers, so the hypothesis holds.  $\square$

## 6 The gap, stated precisely

Corollary 14 is exactly as far as the totally ordered argument reaches, and the obstruction to going further is not a missing estimate but a false statement. The natural strengthening — “ $S_\kappa < S_{\kappa'}$  implies  $W_\kappa \preceq_{\text{lr}} W_{\kappa'}$ ” — is **false**.

**Proposition 15.** *There exist admissible  $\kappa, \kappa'$  with  $S_\kappa < S_{\kappa'}$  and  $W_\kappa \not\preceq_{\text{lr}} W_{\kappa'}$ . For instance at  $n = 5$ ,  $m = 2$ ,  $\mu = (0, 2)$ ,  $\lambda = (1, 5)$ , take  $\kappa = (1, 2)$  and  $\kappa' = (0, 4)$ , so  $S_\kappa = 3 < 4 = S_{\kappa'}$ . Then  $W_\kappa$  is supported on  $\{3, 4, 5, 6\}$  with all values 1, while  $W_{\kappa'}$  is supported on  $\{5\}$  with value 1. At  $(v, v') = (5, 6)$ ,*

$$W_\kappa(5) W_{\kappa'}(6) = 1 \cdot 0 = 0 < 1 = 1 \cdot 1 = W_\kappa(6) W_{\kappa'}(5).$$

Over the range  $n \leq 7$ ,  $m \geq 2$ ,  $d \leq 9$ , the comparable case of Lemma 13 holds on all 15,147 pairs tested, while the incomparable case fails on 692 of 1,652 pairs — a 42% failure rate, not a rare degeneracy. And yet the fibrewise sums satisfy the order on all 327 pairs tested. This localises the gap exactly:

**Gap.** The likelihood ratio order holds between  $W_\kappa$  and  $W_{\kappa'}$  when  $\kappa \leq \kappa'$ , fails in general when  $S_\kappa < S_{\kappa'}$  but  $\kappa, \kappa'$  are incomparable, and is nevertheless restored after summing each fibre  $\{S_\kappa = u\}$ . The missing step is therefore a *cancellation across an antichain*, not a pointwise comparison. No argument that compares individual inner shapes can close it.

Condition (A) is not addressed by this route at all and remains open in the same regime.

## 7 Computational verification

All arithmetic is exact. Eigenvalue counts use the integer characteristic polynomial computed by Faddeev–LeVerrier over  $\mathbb{Q}$  together with Descartes’ rule, which is exact for real-rooted polynomials; no floating point eigenvalues are used for any decision. Controls were re-run this session before the checker was aimed at anything new.

| Control / test                                         | instances       | failures                       |
|--------------------------------------------------------|-----------------|--------------------------------|
| Positive control: $N(s_\lambda)$ Lorentzian (untuned)  | 16              | 0                              |
| Negative control: detector must reject                 | 3               | 3 rejected                     |
| Raw-Hessian lemma (2), symbolic                        | —               | confirmed                      |
| Product-form (Theorem A) control                       | 1105            | 0                              |
| Product-form negative control ( $f_t$ not log-concave) | 5573            | 3604 fail (L3)                 |
| (A) and (B), $\ell = 3, n \leq 8, d \leq 12$           | 77283 Hessians  | 0, 0                           |
| (A) and (B), $\ell = 4, 5, 6, n \leq 6, d \leq 9$      | 143375 Hessians | 0, 0                           |
| (L3) itself over the same ranges                       | 220658 Hessians | 0                              |
| Lemma 3, adversarial integer search                    | 45301           | 0                              |
| negative control, (A) alone                            | 54068           | 21135 have $> 1$ positive eig. |
| negative control, (B) alone                            | 23089           | 526 have $> 1$ positive eig.   |
| Equality criterion of Lemma 3                          | 3921 Hessians   | 0 mismatches                   |
| Interlacing model vs. transfer-matrix DP               | 911             | 0 disagreements                |
| Reformulation of Prop. 9                               | 573             | 0 mismatches                   |
| Lemma 12, random $\text{PF}_2$ quadruples              | 97289           | 0                              |
| negative control, $\text{PF}_2$ dropped                | 484             | 99 fail                        |
| Lemma 13, comparable $\kappa \leq \kappa'$             | 15147 pairs     | 0                              |
| Incomparable $\kappa, S_\kappa < S_{\kappa'}$          | 1652 pairs      | 692 fail                       |
| Fibre sums $F_u$ (this is (B))                         | 327 pairs       | 0                              |

A note on one figure. Of the 3921 Hessians in the  $n \leq 6, d \leq 9, \ell = 3$  range, 2573 (65.6%) have  $\det M = 0$ . An earlier guess that this must be caused by a vanishing diagonal entry was *wrong*: 257 of them have all  $M_{ii} > 0$ , the all-ones matrix among them. The correct characterisation is the one delivered by the multilinear form of the proof — at least two of (B1)–(B3) are equalities — and it matches all 3921 cases.

## 8 Routes killed, with their reasons and the scope of each reason

- **Summand-wise arguments.** Lorentzian polynomials are not closed under addition:  $(x + y)^2 + (2x + y)^2$  has a rank-2 positive semidefinite Hessian. *Scope: none* — this reason carries no hypothesis and kills every route that splits  $s^c$  into a sum and argues termwise. It is what shuts the Schur route even inside Postnikov’s positive range, where the negative-coefficient obstruction does not apply.
- **RLC alone.** Registry `rlc-implies-13`, witness  $\begin{pmatrix} 4 & 6 & 4 \\ 6 & 1 & 4 \\ 4 & 4 & 4 \end{pmatrix}$ . *Scope:  $\ell \geq 3$* , and the witness is  $3 \times 3$ , so it lives in exactly today’s regime. This note identifies the missing hypothesis as (B).
- **The metric criterion beyond  $\ell = 3$ .** Proposition 8. *Scope:  $\ell \geq 4$* . The criterion is a  $3 \times 3$  phenomenon because it is equivalent to the negative-type property of  $e^\phi - 1$ , which no longer follows from the triangle inequality once there are enough points;  $K_{2,3}$  is the obstruction. This does not touch the truth of (L3) at  $\ell \geq 4$ .
- **Ahlswede–Daykin on the interlacing lattice.**  $(S_\kappa, S_\sigma)$  is modular but not a lattice homomorphism, so the four functions theorem gives no pointwise fibre inequality. *Scope: this particular projection*; a different grading of  $\mathcal{L}$  is not excluded.

- **Ordering inner shapes by  $S_\kappa$ .** Proposition 15, 42% failure rate. *Scope:*  $m \geq 2$  with incomparable admissible  $\kappa$ ; at  $m = 1$  the order is total and Corollary 14 goes through.
- **Brändén–Huh closure theorems.** Not used anywhere in this note. The entry for `arXiv:1902.03719` was graded `agent-summary` when this note was written, and the note said reading it at source was the highest-value next step. *Corrected 2026-09-30:* that grade was itself an error. The paper *was* read at source on 2026-09-23, as recorded independently in `reading/2026-09-23-c2.md` l.85 with the fetch mechanics and eight internal labels; the 09-26 downgrade compared a fresh grade against a *stale* note and let the stale note win. The entry is now `deep-read`. What this changes: the definitions and characterisation are citable. What it does not change: no closure theorem is used anywhere in this note, and whether the closure theorems specifically were read on 09-23 is still open, so they remain uncited here. Nothing has been smuggled in as well known.

## 9 What would close it

Two statements, both about the array  $k(a, b) = K_{(a, b, d-a-b)}^c$ , would together prove (L3) at  $\ell = 3$  for every  $m$ , every  $n$  and every shape, by Lemma 3 and Proposition 9:

**Conjecture 16.**  $k(a, b)^2 \geq k(a+1, b) k(a-1, b)$  for all  $a, b$ .

**Conjecture 17.**  $k(a+1, b) k(a, b+1) \geq k(a, b) k(a+1, b+1)$  for all  $a, b$ ; equivalently, the fibre count  $P(u, v)$  of the cyclic interlacing lattice  $\mathcal{L}$  is log-supermodular on  $\mathbb{Z}^2$ .

Conjecture 17 holds when the admissible inner shapes form a chain (Corollary 14). The evidence for both is 220,658 Hessians without an exception, and the data further indicate that both hold at every  $\ell$  tested, not only  $\ell = 3$  — so whatever mechanism produces them is not a  $3 \times 3$  accident, even though the criterion that consumes them is.